Alexander Jacob

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PROFESSIONAL SUMMARY

USC Electrical and Computer Engineering student with hands-on experience in embedded systems, robotics programming, and hardware prototyping. Applied C++, Python, and VEXcode to autonomous and interactive systems, with leadership experience in technical teams and community service.

EDUCATION

University of Southern California — Los Angeles, CA
Bachelor of Science in Electrical and Computer Engineering — Expected May 2030

Shaker High School — Latham, NY
High School Diploma | GPA: 4.0 — Graduated June 2026

EXPERIENCE

FIRST Robotics Team Member — Sep. 2022 - Jun. 2026
Shaker High School, Latham, NY

- Programmed and tested motor controls, sensors, and robot mechanisms in VEXcode as part of a FIRST Robotics team that advanced to the FIRST Championship.
- Developed and tested vision-tracking functionality, debugged robot behavior with programming teammates, and supported iterative integration of software with mechanical systems.

Tech Tutor and Volunteer — Jun. 2022 - Sep. 2025
William K. Sanford Library, Albany, NY

- Provided technology instruction to older adults and community members at the library and local nursing homes.
- Troubleshot Wi-Fi and device issues and recommended computers based on users' needs and budgets.
- Supported literacy programs and community events through setup and visitor assistance.

Front of House Team Member — Apr. 2026 - Jul. 2026
Chick-fil-A, Latham, NY

- Processed orders, prepared beverages and desserts, and maintained dining areas in a high-volume environment.
- Collaborated with team members to maintain speed, accuracy, and guest service during peak periods.

LEADERSHIP

President, Artificial Intelligence Club, Shaker High School — Sep. 2025 - Jun. 2026

- Led 20+ meetings on AI developments, applications, and ethics; managed officers and project leads, coordinated partnerships, and grew active membership to 30.

Eagle Scout, Troop 1277 — 2022 - 2026

- Led the design, construction, and installation of a 16-post Community StoryWalk connecting two local schools.

Student Advisory Board Member, Shaker High School — 2023 - 2026

- Represented student perspectives to school leaders and helped develop a support system for middle school students.

PROJECTS

Mechanical Foosball — 2025 - 2026
VEXcode, VEX V5, Distance Sensors, 3D Printing

- Built a two-player game with 8 buttons, 6 motors, 2 distance sensors, 3D-printed players, and first-to-three scoring.
- Programmed bidirectional motor control, 125 mm goal detection, score tracking, and winner display in VEXcode.

Automated Connect 3 — 2024 - 2025
C++, Arduino Mega, Servos, CAD, 3D Printing

- Designed an accessible 6 x 4 game using an Arduino Mega, 8 micro servos, 6 column-selection buttons, and 24 3D-printed pieces.
- Wrote C++ control logic using a 2D array to validate moves, alternate turns, actuate servos, detect outcomes, and display game state.
- Modeled and 3D-printed 1,230 g of custom components; redesigned ball geometry after testing identified fit issues.

Joystick-Controlled 4-Direction LED PCB — 2026
KiCad, Arduino Uno, PCB Design, Circuit Prototyping

- Built and tested a joystick-controlled 4-direction LED circuit, then created a KiCad schematic and through-hole PCB with routed traces, a GND copper plane, labeled connectors, and ERC/DRC verification.

SKILLS

Languages: C/C++, Python, Java, MATLAB

Hardware: Arduino Mega, VEX V5, servos, distance sensors, motors

Software & Tools: KiCad, Git/GitHub, Arduino IDE, VEXcode, VS Code, Autodesk Inventor, AutoCAD

Technical: PCB design & routing, schematic capture, embedded systems, robotics programming, CAD modeling, 3D printing, testing